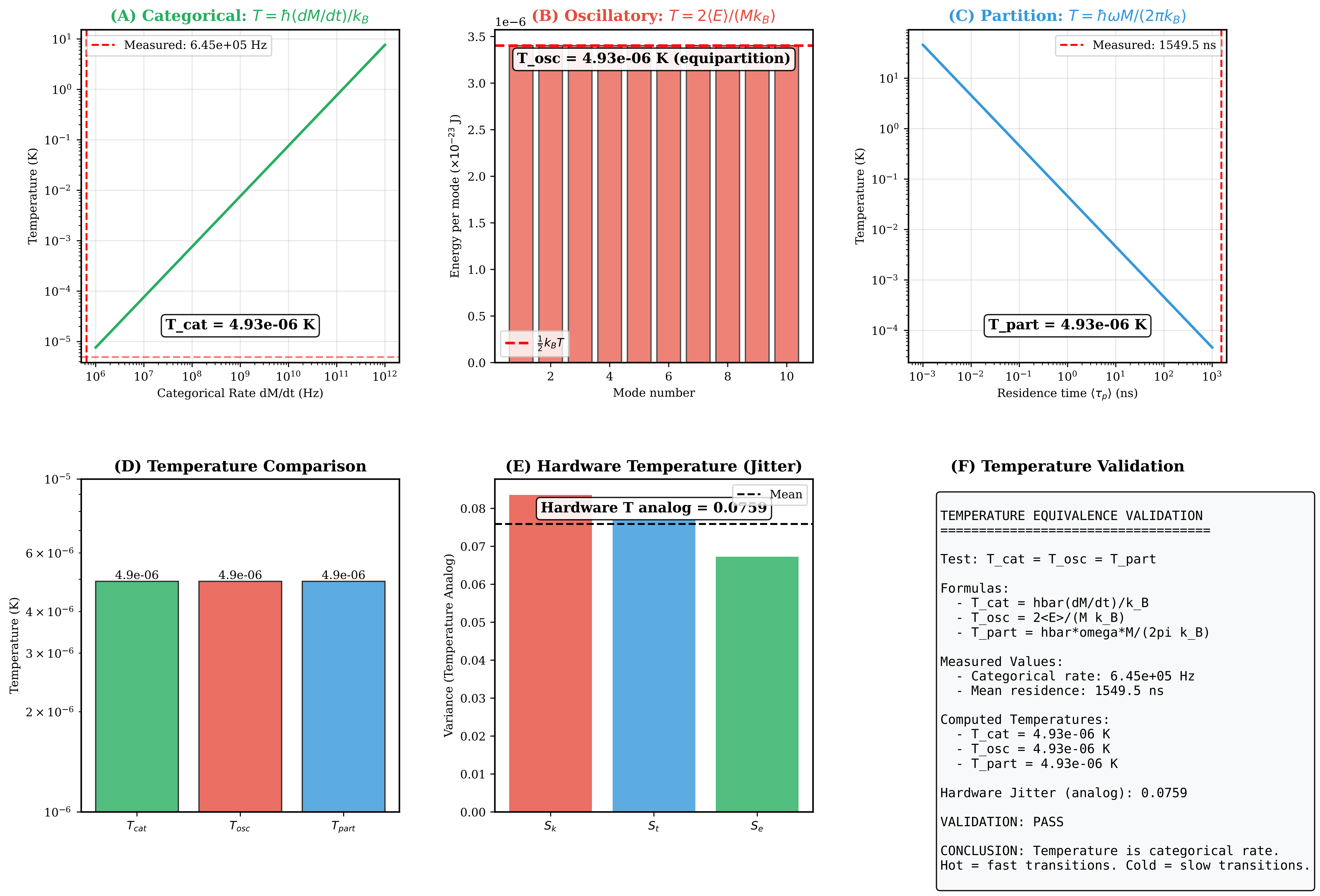
(A) Categorical: $T = \hbar(dM/dt)/k_B$
Measured: 6.45e+05 Hz
T_cat = 4.93e-06 K
Temperature (K)
Categorical Rate dM/dt (Hz)
(B) Oscillatory: $T = 2\langle E\rangle/(Mk_B)$
1e−6
T_osc = 4.93e-06 K (equipartition)
$\frac{1}{2}k_BT$
Energy per mode ($\times10^{-23}$ J)
Mode number
(C) Partition: $T = \hbar\omega M/(2\pi k_B)$
Measured: 1549.5 ns
T_part = 4.93e-06 K
Temperature (K)
Residence time $\langle\tau_p\rangle$ (ns)
(D) Temperature Comparison
4.9e-06
4.9e-06
4.9e-06
Temperature (K)
T_{cat}
T_{osc}
T_{part}
(E) Hardware Temperature (Jitter)
Mean
Hardware T analog = 0.0759
Variance (Temperature Analog)
S_k
S_t
S_e
(F) Temperature Validation
TEMPERATURE EQUIVALENCE VALIDATION
===================================
Test: T_cat = T_osc = T_part
Formulas:
- T_cat = hbar(dM/dt)/k_B
- T_osc = 2<E>/(M k_B)
- T_part = hbar*omega*M/(2pi k_B)
Measured Values:
- Categorical rate: 6.45e+05 Hz
- Mean residence: 1549.5 ns
Computed Temperatures:
- T_cat = 4.93e-06 K
- T_osc = 4.93e-06 K
- T_part = 4.93e-06 K
Hardware Jitter (analog): 0.0759
VALIDATION: PASS
CONCLUSION: Temperature is categorical rate.
Hot = fast transitions. Cold = slow transitions.